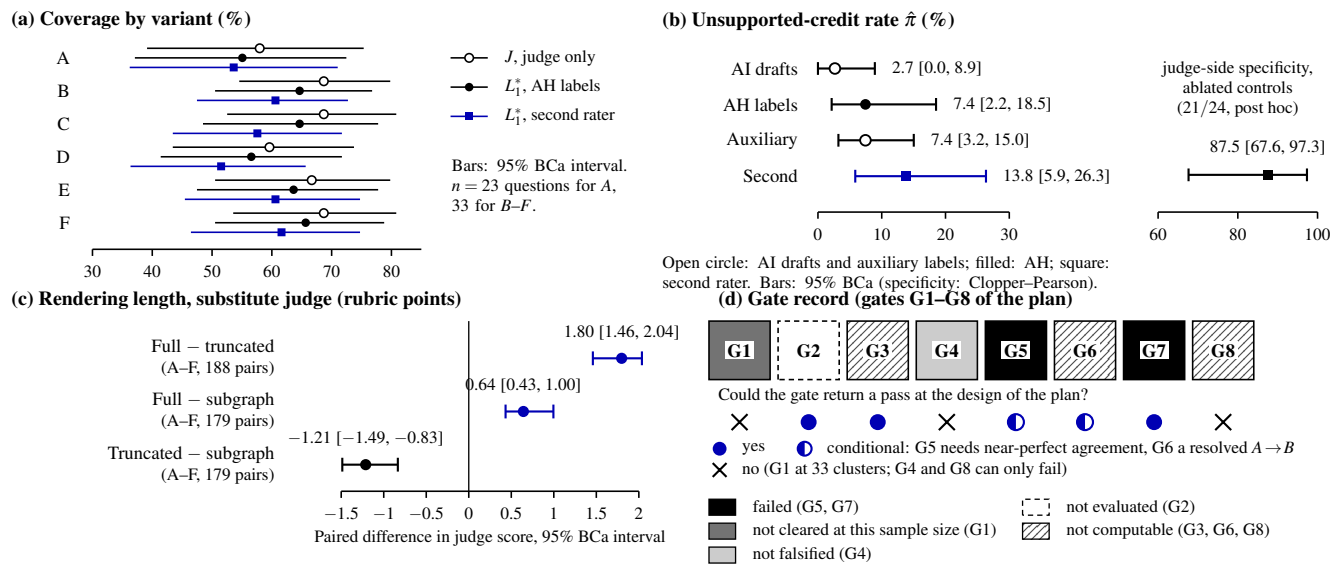

(a) Coverage by variant (%)
A
B
C
D
E
F
30
40
50
60
70
80
J, judge only
L_1^*, AH labels
L_1^*, second rater
Bars: 95% BCa interval.
$n = 23$ questions for A,
33 for B–F.
(b) Unsupported-credit rate $\hat{\pi}$ (%)
AI drafts
2.7 [0.0, 8.9]
AH labels
7.4 [2.2, 18.5]
Auxiliary
7.4 [3.2, 15.0]
Second
13.8 [5.9, 26.3]
0
10
20
30
judge-side specificity,
ablated controls
(21/24, post hoc)
87.5 [67.6, 97.3]
60
80
100
Open circle: AI drafts and auxiliary labels; filled: AH; square:
second rater. Bars: 95% BCa (specificity: Clopper–Pearson).
(c) Rendering length, substitute judge (rubric points)
Full − truncated
(A–F, 188 pairs)
1.80 [1.46, 2.04]
Full − subgraph
(A–F, 179 pairs)
0.64 [0.43, 1.00]
Truncated − subgraph
(A–F, 179 pairs)
−1.21 [−1.49, −0.83]
−1.5
−1
−0.5
0
0.5
1
1.5
2
Paired difference in judge score, 95% BCa interval
(d) Gate record (gates G1–G8 of the plan)
G1
G2
G3
G4
G5
G6
G7
G8
Could the gate return a pass at the design of the plan?
yes
conditional: G5 needs near-perfect agreement, G6 a resolved $A \to B$
no (G1 at 33 clusters; G4 and G8 can only fail)
failed (G5, G7)
not cleared at this sample size (G1)
not falsified (G4)
not evaluated (G2)
not computable (G3, G6, G8)